

Lesson 02: AWS CLI with LocalStack

Learning Objectives

- Install and configure the AWS CLI
 - Configure AWS CLI to work with LocalStack
 - Create, list, and delete AWS resources using the CLI
 - Understand the `--endpoint-url` pattern
 - Verify resources in both CLI and LocalStack Web UI
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1. What is the AWS CLI?

The AWS Command Line Interface (CLI) is a unified tool to manage AWS services from the terminal. Instead of clicking through a web UI, you issue commands like:

```
aws s3 ls
aws dynamodb list-tables
aws sqs list-queues
```

Key insight for this course: We will always use the standard `aws` CLI with the `--endpoint-url` flag pointed at LocalStack. This is the same tool you'd use against real AWS — no special wrappers.

2. Install AWS CLI

macOS

```
# Using Homebrew
brew install awscli

# Or using the official installer
curl "https://awscli.amazonaws.com/AWSCLIV2.pkg" -o "AWSCLIV2.pkg"
sudo installer -pkg AWSCLIV2.pkg -target /
```

Linux (Debian/Ubuntu)

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o
"awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
```

Windows

Download the installer from: <https://awscli.amazonaws.com/AWSCLIV2.msi>

Verify Installation

```
aws --version
```

Expected output: `aws-cli/2.x.x ...`

3. Configure AWS CLI for LocalStack

AWS CLI stores credentials and region in `~/.aws/config` and `~/.aws/credentials`. LocalStack accepts any credentials — we use dummy values.

```
aws configure
```

Enter the following when prompted:

```
AWS Access Key ID [None]: test
AWS Secret Access Key [None]: test
Default region name [None]: us-east-1
Default output format [None]: json
```

This creates two files:

`~/.aws/credentials`:

```
[default]
aws_access_key_id = test
aws_secret_access_key = test
```

`~/.aws/config`:

```
[default]
region = us-east-1
output = json
```

Why `test/test`? LocalStack doesn't validate credentials. Any value works. Real AWS would require valid credentials from the IAM service.

4. The `--endpoint-url` Pattern

By default, `aws` CLI commands connect to the real AWS service endpoints (e.g., `s3.amazonaws.com`). To point them at LocalStack, add:

```
--endpoint-url=http://localhost:4566
```

Every command in this course follows this pattern:

```
aws --endpoint-url=http://localhost:4566 <service> <command>
```

Creating an Alias (Optional)

To save typing, some developers create an alias:

```
alias awslocal='aws --endpoint-url=http://localhost:4566'
```

Then use:

```
awslocal s3 ls
```

However, for this course we will write the full `--endpoint-url` in documentation so you see the complete pattern. You may use the alias in your own terminal.

5. Managing Resources via CLI

Make sure LocalStack is running:

```
docker ps | grep localstack
```

S3 — Create a Bucket

```
aws --endpoint-url=http://localhost:4566 s3 mb s3://cli-bucket
```

Expected output:

```
make_bucket: cli-bucket
```

List buckets:

```
aws --endpoint-url=http://localhost:4566 s3 ls
```

Upload a file:

```
echo "hello from cli" > sample.txt  
aws --endpoint-url=http://localhost:4566 s3 cp sample.txt s3://cli-bucket/
```

List objects:

```
aws --endpoint-url=http://localhost:4566 s3 ls s3://cli-bucket/
```

DynamoDB — Create a Table

```
aws --endpoint-url=http://localhost:4566 dynamodb create-table \  
  --table-name CliTable \  
  --key-schema AttributeName=id,KeyType=HASH \  
  --attribute-definitions AttributeName=id,AttributeType=S \  
  --billing-mode PAY_PER_REQUEST
```

List tables:

```
aws --endpoint-url=http://localhost:4566 dynamodb list-tables
```

Put an item:

```
aws --endpoint-url=http://localhost:4566 dynamodb put-item \  
  --table-name CliTable \  
  --item '{"id": {"S": "1"}, "name": {"S": "Alice"}, "role": {"S":  
"engineer"}}'
```

Scan the table:

```
aws --endpoint-url=http://localhost:4566 dynamodb scan \  
  --table-name CliTable
```

q

SQS — Create a Queue

```
aws --endpoint-url=http://localhost:4566 sqs create-queue \  
  --queue-name cli-queue
```

Expected output:

```
{  
  "QueueUrl": "http://sqs.us-east-  
1.localhost.localstack.cloud:4566/000000000000/cli-queue"  
}
```

List queues:

```
aws --endpoint-url=http://localhost:4566 sqs list-queues
```

Send a message:

```
aws --endpoint-url=http://localhost:4566 sqs send-message \  
  --queue-url http://sqs.us-east-  
1.localhost.localstack.cloud:4566/000000000000/cli-queue \  
  --message-body "Hello from CLI"
```

Receive a message:

```
aws --endpoint-url=http://localhost:4566 sqs receive-message \  
  --queue-url http://sqs.us-east-  
1.localhost.localstack.cloud:4566/000000000000/cli-queue
```

6. Deleting Resources

S3 — Delete Objects and Bucket

```
aws --endpoint-url=http://localhost:4566 s3 rm s3://cli-bucket/sample.txt  
aws --endpoint-url=http://localhost:4566 s3 rb s3://cli-bucket
```

DynamoDB — Delete Table

```
aws --endpoint-url=http://localhost:4566 dynamodb delete-table \  
  --table-name cli-table
```

```
--table-name CliTable
```

SQS — Delete Queue

```
aws --endpoint-url=http://localhost:4566 sqs delete-queue \
  --queue-url http://sqs.us-east-
1.localhost.localstack.cloud:4566/000000000000/cli-queue
```

7. Verifying in the Web UI

After creating resources via CLI, open the LocalStack Web UI at:

```
http://localhost.localstack.cloud:4566
```

Navigate to each service:

Service	What to verify
S3	Bucket cli-bucket appears, sample.txt listed inside
DynamoDB	Table CliTable appears with status ACTIVE , item count 1
SQS	Queue cli-queue appears with URL matching CLI output

This confirms that CLI-created resources are visible to the Web UI and vice versa — they share the same LocalStack instance.

8. Key Takeaways

- **--endpoint-url=http://localhost:4566** is the universal pattern for using AWS CLI with LocalStack
- The same CLI commands work against real AWS — only the endpoint changes
- All three services (S3, DynamoDB, SQS) have identical CLI APIs to real AWS
- Resources created via CLI are immediately visible in the Web UI
- The alias **awslocal** is optional; we use the full command for clarity

example and exercise repo url

<https://github.com/eyu-se/terraform-training-material-with-projects>

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